Pochimireddy Ajith Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.08
Sri Chaithanya	Class XII (BIEAP)	2023	93.50%
Sri Chaithanya	Class X (SSC)	2021	100.00%

WORK EXPERIENCE

AI/ML Intern | Country Delight

- Retrained and fine-tuned a **YOLO object-detection model** for **Proof-of-Delivery (POD) image analysis** using dataset cleaning, targeted augmentation, and hyperparameter tuning, improving **validation mAP@50 from 74% to 82%**.
- Developed an **internal computer-vision model-training tool** supporting **image-classification and object-detection workflows**, enabling the team to conduct repeatable experiments across datasets and model configurations.
- Redesigned an **MCP-based chatbot** with intent-driven routing to business APIs or smaller language models, using **confidence-gated LLM fallback** to reduce unnecessary high-cost model calls.
- Trained **lightweight intent-classification models** on user queries traced through **Langfuse** and implemented **caching for repeated informational queries**.

Research Intern - Graph ML for Gas-Pipeline Reliability | IIT Hyderabad

- Developed **GraphSAGE and Graph Attention Network (GAT)** transfer-learning models for **seismic reliability estimation of gas-pipeline networks** under the guidance of Dr. Sourav Das.
- Pretrained the models on approximately **1,200 source-network scenarios** and fine-tuned them on approximately **60 target-network scenarios**, achieving **R^2 = 0.972, MAE = 0.015, and MSE = 0.0014** on the target network.
- Extending the framework toward **reliability-aware repair prioritization using reinforcement learning**.

PROJECTS

CaseFlow -- Full Stack Incident Management Platform

React, TypeScript, Django REST Framework, PostgreSQL, JWT, Docker, AWS, GitHub Actions

- Built an **end-to-end incident management platform** with React/TypeScript and Django REST APIs, supporting incident reporting, search, comments, role-based dashboards, and a **5-state workflow**.
- Implemented **JWT authentication, role- and object-level authorization, audit logging, and workflow validation**, preventing unauthorized access and invalid state transitions.
- Developed responsive frontend components with API-driven state management, filtering, form validation, loading/error states, and reusable UI components.
- Containerized and deployed the application using **Docker and AWS**, with automated testing and CI/CD through **GitHub Actions**.

FlowForge -- Data Workflow & SQL Assistant

Python, FastAPI, PostgreSQL/MySQL/Redshift, SQLAlchemy, Pandas, APScheduler, SQLGlot, LLM API

- Built a **backend workflow platform** to extract, transform, and load data across relational databases, with configurable jobs, scheduling, retries, and execution logs.
- Implemented **incremental checkpoints and failure recovery** to avoid duplicate processing and enable reliable reruns of interrupted workflows.
- Developed an **NL-to-SQL assistant** that converts user questions into SQL and validates generated queries using **SQLGlot AST analysis** before execution.
- Designed REST APIs and database abstractions for managing workflows, connections, run history, and query execution across multiple data sources.

DocuLens--AI-Powered Document Intelligence Platform

React, FastAPI, Sentence Transformers, FAISS, PyMuPDF, LLM API

- Built a **full-stack document intelligence application** enabling users to upload documents, ask natural-language questions, and receive grounded answers with page-level source citations.
- Developed a **React conversational interface** with document upload, processing states, chat history, citation navigation, and API-based interaction with the FastAPI backend.
- Implemented a RAG pipeline using **page-aware extraction, semantic chunking, Sentence Transformer embeddings, and FAISS retrieval** to provide contextually relevant responses.
- Added retrieval thresholds, timeouts, logging, and offline evaluation using **Recall@K and citation accuracy** to improve reliability and answer quality.

SKILLS

Programming & Backend: Python, SQL, FastAPI, Django REST Framework, REST APIs

Databases & Data Engineering: PostgreSQL, MySQL, SQLAlchemy, Pandas, ETL/ELT, data validation

Machine Learning & Computer Vision: PyTorch, scikit-learn, YOLO, image classification, object detection, GraphSAGE, Graph Attention Networks, model evaluation

LLMs & Generative AI: Hugging Face Transformers, LLMs/SLMs, RAG, embeddings, semantic retrieval, grounded generation, prompt evaluation

Tools & Frameworks: MCP, Langfuse, FAISS, SQLGlot, APScheduler, PyMuPDF, Pytest, Git/GitHub, Linux

Engineering Practices: API design, authentication and authorization, automated testing, secure query validation